

Fehmi Özbayrak

Curriculum Vitae · Ph.D. Student in Petroleum and Geosystems Engineering

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Research Profile

Computational energy researcher connecting reservoir engineering and geomechanics, scientific software, spatial machine learning, uncertainty quantification, and techno-economic analysis. Research spans produced-water disposal and induced seismicity, geological poromechanics, geothermal development, and data-center energy systems.

Education

Ph.D. Student, Petroleum and Geosystems Engineering — The University of Texas at Austin

Austin, Texas · Expected Aug 2027

Advisor: Dr. Mukul M. Sharma. Tentative dissertation: *Multiscale Geomechanics of Produced-Water Disposal in the Delaware Basin: Fracture Containment, Ground Deformation, and Induced Seismicity*. Cumulative UT M.S./Ph.D. GPA: 3.61/4.00.

MSc, Petroleum and Geosystems Engineering — The University of Texas at Austin

Austin, Texas · May 2024

Thesis: *A novel spatial bagging algorithm for predictive ensemble machine learning*. Co-advisors: Dr. John T. Foster and Dr. Michael J. Pyrcz.

BEng, Petroleum and Natural Gas Engineering — Middle East Technical University

Ankara, Türkiye · Feb 2022

GPA: 3.21/4.00.

Research and Professional Experience

Graduate Research Assistant — UT Austin / Center for Integrated Seismicity Research

Austin, Texas · Summer 2024–Present

- Build and calibrate coupled reservoir-geomechanics models using well logs and injection and pressure histories to investigate fracture containment, pressure transmission, surface deformation, and induced-seismicity risk from produced-water disposal.
- Design sensitivity studies that separate stress, operating, pressure, and leakoff controls while communicating calibration non-uniqueness and model limitations.
- Develop auditable geological-poromechanics workflows and verification cases for MOOSE/PorousFlow simulation.

Reservoir Engineering Intern — Fervo Energy

Houston, Texas · May–Aug 2025

- Architected and implemented a modular Python platform for geothermal development planning, connecting configurable well schedules and surface-facility allocation to daily power forecasts, costs, and contract economics.
- Integrated a machine-learning proxy for production-temperature evolution to accelerate repeated schedule evaluation.
- Formulated drilling start times as Pyomo optimization variables and maximized project revenue subject to power-delivery constraints; built interactive Plotly and Dash analysis tools.

Graduate Research Assistant — DiReCT Consortium, The University of Texas at Austin

Austin, Texas · 2022–2024

- Developed Spatial Bagging, an ensemble-learning method that uses geostatistical effective sample size to incorporate spatial dependence into bootstrap sampling.
- Led software, experiments, and first-author writing for two peer-reviewed papers on spatial prediction and uncertainty quantification; coauthored a later Spatial Random Forest extension.

Drilling Engineer Intern — Middle East Air Drilling Services

Alaşehir, Türkiye · 2021

- Completed geothermal drilling field training around a ZJ-40 double mobile rig, performing mud and cement calculations and formation-evaluation exercises while documenting casing, cementing, MWD, and coiled-tubing operations.

Undergraduate Researcher — Middle East Technical University

Ankara, Türkiye · Poster presented Dec 2021

- Created synthetic permanent-downhole-gauge data, introduced noise and outliers, and compared nonlinear regression and wavelet-based denoising methods; presented the work at the ADIMODTÜ Project Poster Presentation.

Research Software and Selected Research Projects

JAVELINA — Auditable Geological Poromechanics Workflow 2026

- Developed a declarative Python workflow that converts geological surfaces, meshes, material data, and well schedules into auditable MOOSE/PorousFlow cases for serial and MPI execution.
- Implemented completion intersection, conservative signed-source allocation, provenance tracking, and automated result-quality reporting.
- Verified generated operators against six analytical, semi-analytical, and manufactured benchmarks, with near-second-order spatial convergence and 1.196% maximum peak-normalized error. The project remains unpublished.

Data Center Integrated Systems Model (DC-ISM) 2025–2026

- Built a modular, configuration-driven Python demand-side engine that maps AI workload states and hardware fleets to hourly IT, cooling, electrical, water, PUE, cost, carbon, and flexibility profiles.
- Implemented measured-weather ingestion, tariffs, sensitivity screening, correlated uncertainty propagation, scenario-year bootstrapping, supply-land screening, and risk-aware design comparison.
- Defined typed interfaces, explicit scope boundaries, a limitations register, and a 155-test suite to keep assumptions and unsupported effects auditable. The project remains private.

Spatial Bagging for Prediction and Uncertainty Quantification 2022–2025

- Developed an ensemble-learning algorithm that adjusts bootstrap sample size using geostatistical effective sample size so spatial dependence is represented rather than treated as independent data.
- Evaluated prediction and uncertainty behavior across synthetic two-dimensional datasets spanning spatial-correlation and noise regimes; released the [public Python implementation](#) and published two first-author papers.

Delaware Mountain Group Fracture-Containment Study 2024–Present

- Built and calibrated a coupled flow-and-fracture model against multi-year injection history and daily pressure observations from Delaware Basin disposal wells.
- Designed sensitivity studies across stress stratigraphy, injection rate and intermittency, pressure, and leakoff to distinguish controls on vertical containment, pressure buildup, and lateral fracture growth. Manuscript in preparation.

Sierra Solutions — Gas-to-Data-Center Investment Model 2026

- Built a 15-year physics-and-finance model for a 15 MW firm / 20.25 MW installed behind-the-meter natural-gas power concept serving an AI data center, integrating redundancy, fuel logistics, contracts, debt, and project economics.
- Evaluated 10,000 Monte Carlo scenarios across 12 risk drivers and presented a base case with 14.47% levered IRR, \$2.32 million NPV at 12%, and 1.72x minimum DSCR; earned second place in the North American SPE Energython as a solo entrant.

Technical Competition Projects

Utah FORGE — Engineering Analysis, GEODE Engineering Datathon 2nd place · 2025 · Two-person team

- Wrote the analysis code integrating tracer, distributed fiber-optic temperature, strain and acoustic data, microseismicity, well surveys, and stimulation records to infer stage-level connectivity and preferential flow paths.
- Proposed a targeted restimulation program using perturbed-stress interpretation, oriented perforations, isolation and diversion, and pulsed initiation; recommendations were not represented as implemented results.

Nevada Blind-Geothermal Exploration — GEODE Geosciences Datathon 2nd place · 2025 · Two-person team

- Built a spatial CNN workflow using strain, gravity, permeability, modeled temperature, earthquake-density, and fault-distance rasters to reproduce and interrogate play-fairway favorability patterns.
- Used geographically separated development and validation regions to identify candidate zones for follow-up without presenting them as physically validated discoveries.

Hydraulic-Fracturing Fuel Forecast — UT PGE Energy AI Hackathon 1st place · 2025 · Five-person team

- Served as the main machine-learning architect and built most of a random-forest regression workflow forecasting grid, diesel, and CNG requirements from operational, fleet, fuel, formation, and weather features.
- Compared 12 CatBoost-derived features with 1,082 one-hot features and calibrated ensemble uncertainty across 100 repeated runs.

Missing-Well Production Forecasting — UT Energy AI Hackathon Best Presentation · 2026 · Team Chain

- Served as the main machine-learning engineer for a field-scale workflow predicting oil, water, and gas production for wells omitted from the supplied dataset.

Utah FORGE Microseismic Analysis — SPE Geothermal Datathon 2nd place · 2023 · Four-person team

- Applied bootstrap ensemble clustering to microseismic events, clustering each resampled realization and aggregating the results to interpret event structure.

Peer-Reviewed Publications

1. **Özbayrak, F.**, Foster, J. T., and Pyrcz, M. J. (2025). “Spatial bagging for predictive machine learning uncertainty quantification.” *Computers & Geosciences*, 203, 105947. doi:10.1016/j.cageo.2025.105947. First and corresponding author.
2. Merzoug, A., **Özbayrak, F.**, Foster, J. T., and Pyrcz, M. J. (2025). “Beyond random forest: how spatial bagging and spatial random forest dominate for subsurface applications?” *Computational Geosciences*, 29, article 52. doi:10.1007/s10596-025-10388-0.
3. **Özbayrak, F.**, Foster, J. T., and Pyrcz, M. J. (2024). “Spatial bagging to integrate spatial correlation into ensemble machine learning.” *Computers & Geosciences*, 186, 105558. doi:10.1016/j.cageo.2024.105558. First and corresponding author.

Manuscripts in Preparation

- **Özbayrak, F.** “JAVELINA: an accessible, quality-transparent workflow for geological poromechanics with MOOSE.” Manuscript in preparation; not submitted.
- **Özbayrak, F.** et al. “Induced fracture geometry and vertical containment during shallow wastewater injection in the Delaware Mountain Group: A calibrated geomechanical modeling study.” Manuscript in preparation; title and author order subject to coauthor review.

Awards and Honors

- Second Place, North America SPE Asset Management Technical Section Energython — Sierra Solutions, solo entry, 2026.
- Judges’ Selection Award for Best Presentation, UT Energy AI Hackathon — Team Chain, 2026.
- First Place, UT PGE Energy AI Hackathon — Data Comrades, five-person team, 2025.
- Second Place, GEODE Engineering Datathon — Operation Hotspot, two-person team, 2025.
- Second Place, GEODE Geosciences Datathon — Operation Hotspot, two-person team, 2025.
- Second Place, SPE Geothermal Datathon — Team NDT, four-person team, 2023.

Selected Technical Presentations

- “Sierra Solutions,” North America SPE Asset Management Technical Section Energython Final, Jan 2026.
- Utah FORGE engineering analysis and Nevada blind-geothermal exploration, GEODE Datathon, 2025.
- Hydraulic-fracturing fuel forecasting, UT PGE Energy AI Hackathon, 2025.
- **Özbayrak, F.** and Gökalp Yavuz, F. “Denoising and Outlier Removal from Permanent Downhole Gauge Using Nonlinear Regression Methods,” ADIMODTÜ Project Poster Presentation, 2021. [hdl:11511/97232](https://hdl.handle.net/11511/97232).

Teaching

Teaching Assistant, PGE 383/379 — High Performance Computing for Engineers — The University of Texas at Austin

Austin, Texas · Spring 2024

Supported Julia programming laboratories and occasional office hours for approximately 30 students in Dr. John T. Foster’s course.

Leadership and Service

President — Switch Energy Alliance UT Chapter

Austin, Texas · 2026–Present

Lead an eight-officer student organization and secured an approximately \$3,000 budget while building undergraduate representation and cross-campus energy partnerships.

Vice President for External Affairs — Switch Energy Alliance UT Chapter

Austin, Texas · 2024–2025

Challenge Co-Architect — Fourth Annual UT PGE Energy AI Hackathon

Austin, Texas · 2024 event cycle

Co-architected the challenge with Elnara Rustamzade and supported teams and event logistics for approximately 160 students across 25 teams.

Technical Skills

Programming and software: Python, C/C++, Julia, NumPy, Pandas, scikit-learn, Pyomo, Plotly, Dash, Git, Linux/shell, MPI, LaTeX.

Subsurface and computational methods: Reservoir simulation, coupled flow-and-fracture modeling, finite-element poromechanics, MOOSE/PorousFlow, ResFrac, history matching, calibration, numerical verification, sensitivity analysis.

Data and uncertainty: Geostatistics, spatial statistics, ensemble machine learning, random forests, raster/GIS workflows, uncertainty quantification, Monte Carlo analysis, tracer/fiber-optic/microseismic interpretation.

Energy systems: Geothermal systems, techno-economic modeling, project finance, mathematical optimization, hourly facility-demand modeling, data-center cooling, water, PUE, tariffs, and siting analysis.